

Student 23 **MALLI CHANDANA** 192472250RD

5 / 5 submitted

Q1. Exception**Score: 10.00** **TC: 3/3 passed****CODE**

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr = {10, 20, 30, 40, 50};

        try {
            int index = sc.nextInt();

            if (index < 0 || index >= arr.length) {
                throw new ArrayIndexOutOfBoundsException();
            }

            System.out.println("Element at index " + index + ": " + arr[index]);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Exception: Invalid array index.");
        }
        finally {
            sc.close();
        }
    }
}
```

INPUT

6

OUTPUT

Exception: Invalid array index.

Q2. IllegalArgumentException

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int mark = sc.nextInt();

        try {
            if (mark < 0 || mark > 100) {
                throw new IllegalArgumentException("Invalid marks. Marks should be between 0 and 100.");
            }
            System.out.println("Valid marks: " + mark);
        } catch (IllegalArgumentException e) {
            System.out.println("Exception: " + e.getMessage());
        }
    }
}
```

INPUT

120

OUTPUT

Exception: Invalid marks. Marks should be between 0 and 100.

Q3. user-defined exception named WeakPasswordException.

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String password = sc.nextLine();

        try {
            if (password.length() < 8) {
                throw new WeakPasswordException(
                    "Exception: Password must contain at least 8 characters."
                );
            }

            System.out.println("Password is strong.");
        }
        catch (WeakPasswordException e) {
            System.out.println(e.getMessage());
        }

        sc.close();
    }
}

class WeakPasswordException extends Exception {
    public WeakPasswordException(String message) {
        super(message);
    }
}
```

INPUT

java123

OUTPUT

Exception: Password must contain at least 8 characters.

Q4. Email Validation

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String email = sc.nextLine();

        try {
            validateEmail(email);
        }
        catch (InvalidEmailException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        sc.close();
    }

    public static void validateEmail(String email)
        throws InvalidEmailException {

        if (!email.contains("@")) {
            throw new InvalidEmailException("Invalid email address.");
        }
        else {
            System.out.println("Email is valid.");
        }
    }
}

class InvalidEmailException extends Exception {
    public InvalidEmailException(String message) {
        super(message);
    }
}
```

INPUT

student@gmail.com

OUTPUT

Exception: Invalid email address.

Q5. Banking Transaction Using Exception Propagation

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;

public class Main {

    static int balance;

    static void withdraw(int amount) throws InsufficientBalanceException {
        validateBalance(amount);
    }

    static void validateBalance(int amount) throws InsufficientBalanceException {
        if (amount > balance) {
            throw new InsufficientBalanceException(
                "Withdrawal amount exceeds available balance."
            );
        }
    }

    static void processTransaction(int amount) throws InsufficientBalanceException {
        withdraw(amount);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        sc.nextLine(); // reads "Balance:"
        balance = sc.nextInt();

        sc.nextLine(); // clears remaining line
        sc.nextLine(); // reads "Withdraw:"
        int amount = sc.nextInt();

        try {
            processTransaction(amount);
        }
        catch (InsufficientBalanceException e) {
            System.out.println("Exception propagated to main().");
            System.out.println("InsufficientBalanceException: " + e.getMessage());
        }
    }

    class InsufficientBalanceException extends Exception {

        InsufficientBalanceException(String message) {
            super(message);
        }
    }
}
```

INPUT

```
Balance:
5000
Withdraw:
7000
```

OUTPUT

```
Exception propagated to main().
InsufficientBalanceException: Withdrawal amount exceeds available balance.
```